

Trading the Forward Bias: Are there Limits to Speculation?*

MARKUS HOCHRADL

Vienna Graduate School of Finance
Nordbergstraße 15, 1090 Vienna, Austria

CHRISTIAN WAGNER[†]

Department of Finance and Accounting
Vienna University of Economics and Business Administration
Nordbergstraße 15, 1090 Vienna, Austria

Abstract

In this paper we investigate whether trading the forward bias allows for economically significant excess returns. We find that bias-trading strategies can be viewed as attractive investment opportunities per se, useful diversification devices, and promising portfolio extensions for active fund managers trying to beat their benchmarks. The empirical results, which also mirror the problems arising in attempts to explain the puzzle by risk-premia, are consistent with market evidence that the bias is traded in practice. Overall, our findings suggest that limits to speculation are unlikely to provide a (stand-alone) explanation for the persistence of the forward bias.

Keywords: Exchange rates, forward bias, trading strategies, currency options, limits to speculation.

JEL classification: F31

*This paper was partly written while Markus Hochradl was visiting the Department of Finance at the NYU Leonard N. Stern School of Business and while Christian Wagner was a research fellow at the Vienna Graduate School of Finance. The stimulating research environment is gratefully acknowledged. We thank Tom Berglund, Thomas Dangl, Engelbert Dockner, Alois Geyer, Antti Ilmanen, Robert Jung, Thomas Nagel, Vasant Naik, Otto Randl, and Josef Zechner as well as the seminar participants at the University of Vienna, the Institute of Strategic Capital Market Research, the Tanaka Business School at the Imperial College London, and the 2006 Meeting of the European Finance Association for their helpful comments and suggestions. Any remaining errors are our own.

[†]Corresponding author. Phone: +43-1-31336-4253; Fax: +43-1-31336-736; E-mail: christian.wagner@wu-wien.ac.at.

1 Introduction

Uncovered interest rate parity (UIP) postulates that the expected change in the exchange rate compensates for the interest rate differential or, given that covered interest rate parity holds, equals the forward premium. Following the work by Bilson (1981) and Fama (1984), an enormous amount of research on testing whether the forward rate is an unbiased predictor of the future spot rate has accumulated over the last decades. Surveys of the literature include Hodrick (1987), Froot and Thaler (1990), Lewis (1995), Engel (1996), and Sarno (2005). Although results vary depending on how exchange rates are modeled, the common finding in the overwhelming majority of past research is that the forward rate is not an unbiased predictor of the future spot rate. So far, this forward bias has not been convincingly explained and continues to baffle the international finance community.

The concept of limits to speculation as put forward by Lyons (2001) suggests that the forward bias - although it might be statistically significant - is economically small. In particular, he argues that the bias is too small to attract speculative capital, since other investment opportunities yield higher Sharpe ratios. As a consequence, the bias remains unexploited and (statistically) persistent. Recent work inspired by the limits to speculation hypothesis finds that exchange rate dynamics are characterized by a time-varying forward bias. Sarno et al. (2006), for instance, argue that a statistically persistent but economically small forward bias exists most of the time. Market evidence, however, suggests that the forward bias is traded and that bias-trading might even be a key driver for the surge in foreign exchange trading over the last years; see e.g. Galati and Melvin (2004) and Galati et al. (2007). Research and strategy reports published by financial institutions clearly show that speculative capital is allocated to bias-trading strategies. Although financial institutions usually do not fully disclose their approaches, the fact that bias-exploiting strategies have been used for years and that their use continues and even gets more widespread suggests (a certain degree of) economic significance.

In this paper we assess the concept of limits to speculation empirically. We develop implementable trading strategies explicitly aimed at exploiting the forward bias in bilateral and multi-currency settings. In particular, we use optimal carry-trade portfolios as well as approaches based on currency options and investigate whether these strategies allow for economically significant excess returns and thus should have the potential to attract speculative capital.

Our results indicate that the forward bias does not only exist statistically but that betting against UIP yields Sharpe ratios higher than those of the S&P 500 Index, the MSCI World Total Return Index, and a JP Morgan US Government Bond Index, even after adjusting for transaction costs. Bias-trading strategies can be viewed as attractive investment opportunities per se (higher Sharpe ratios), useful diversification devices (no systematic risk), and promising portfolio extensions for active fund managers trying to outperform their benchmarks (higher Sharpe ratios and positive Jensen's alphas).

Overall, our results challenge the limits to speculation hypothesis but are in line with market evidence that the bias is traded. Furthermore, our findings mirror the problems arising in attempts to explain the puzzle by risk premium arguments.

The remainder of the paper is organized as follows. Section 2 contains a review of the related literature on UIP and limits to speculation as well as a discussion on market evidence on bias-trading. We describe our approaches to trade the bias using optimal carry-trade portfolios and currency options strategies in Section 3. In Section 4 we report the empirical results which we interpret in the context of limits to speculation and attempts to explain the bias by risk-premia. Section 5 summarizes our findings and concludes. In the Appendix we provide (A) a discussion on the implications of nonlinearity between spot and forward rates for bias-trading, (B) the specification of the error correction model, and (C) an outline of the heuristic portfolio optimization algorithm.

2 Uncovered Interest Rate Parity, Limits to Speculation, and Market Evidence

UIP postulates that the expected change in the exchange rate equals the interest rate differential or, given that covered interest rate parity holds, the forward premium,

$$\Delta s_{t+k}^e = i_{t,k} - i_{t,k}^* = f_{t,k} - s_t, \quad (1)$$

with $\Delta s_{t+k} \equiv s_{t+k} - s_t$, where the superscript e denotes the expectation based on information at time t , s_t and $f_{t,k}$ denote the natural logarithm of the spot rate, S_t (given as the domestic price per unit of foreign currency), and the k -period forward rate, $F_{t,k}$, and $i_{t,k}$ and $i_{t,k}^*$ the effective k -period domestic and foreign interest rates.

Tests of UIP are frequently based on the Fama regression, following Fama (1984), which, motivated by Eq. (1), relates the change in the spot rate to a constant and the forward premium,

$$\Delta s_{t+1} = \alpha + \beta (f_{t,1} - s_t) + u_{t+1}, \quad (2)$$

where, for simplicity, $k = 1$ and u_{t+1} denotes a disturbance term. The hypothesis of UIP being valid is that the constant (α) should equal zero, the slope coefficient (β) unity, and the disturbance term should be serially uncorrelated. The common finding, however, is that the slope coefficient is less than unity, indicating that the higher interest rate currency tends to not depreciate as much as predicted by UIP, and often negative, indicating not only that UIP does not hold but also that the higher interest rate currency tends to appreciate rather than depreciate. So far, attempts to explain the forward bias

by a time-varying risk premium, bubbles, learning, and peso problems, have met with limited success. For surveys of the literature see e.g. Hodrick (1987), Froot and Thaler (1990), Lewis (1995), Engel (1996), and Sarno (2005).

While the forward bias implies the predictability of excess returns over UIP, Lyons (2001) claims that the profitability of related trading strategies is limited. In particular, his concept of limits to speculation postulates that deviations from UIP persist because nobody is willing to trade on these deviations, since other investment opportunities yield higher Sharpe ratios. Thus, even if the forward bias is significant statistically, the failure of UIP might not be substantial in economic terms: deviations from UIP might be too small to attract trader's capital, who consequently leave the bias unexploited and (statistically) persistent.¹ Lyons (2001) stresses that financial institutions typically allocate capital based on Sharpe ratios and that it is this empirical fact which is essential to the limits to speculation hypothesis rather than a theoretical rationale for why this behavior arises. Given that the annual Sharpe ratio for a buy-and-hold equity strategy in the US market has averaged at about 0.4 over the last 50 years, this figure can be considered as a lower bound for traders' Sharpe ratio threshold determining whether they take up a bias-trading strategy or not.² However, referring to interviews with practitioners, Lyons (2001) reports that traders' interest would be limited to trading strategies with Sharpe ratios above 0.5. Analyzing bias trading strategies across six USD exchange rates, he finds Sharpe ratios (after approximated transaction costs) between 0.37 and 0.41 and thus support for his hypothesis.

Villanueva (2007) finds that the forward premium has directional predictability which translates into statistically significant profits from trading on the forward bias based. Sarno et al. (2006) investigate the relation between spot and forward rates in an exponential smooth transition regression (STR) framework inspired by limits to speculation. They find that large (expected) deviations from UIP are followed by reversions towards UIP, whereas small deviations lead to a statistically persistent but economically unimportant forward bias.³ The work of Burnside et al. (2006) also seems to be consistent with limits to speculation, since they argue that transaction costs and price pressure limit the extent to which traders try to exploit the anomaly.

In the model of Plantin and Shin (2007) carry-trades create self-enforcing speculation opportunities. This corresponds to the practitioner's view that the forward bias is not only necessary for carry trades but also driven by them. Galati and Melvin (2004) and Galati et al. (2007) provide evidence that

¹This hypothesis builds on the relation between order flow and price adjustment. As noted in Lyons (2001), unless all information relevant for exchange rates and the mapping from that information to prices are publicly known, order flow will influence price adjustments. The latter requirement is obviously violated but even the former is unlikely to be satisfied; see Lyons (2001) and Evans and Lyons (2002).

²The figure of 0.4 is reported by Lyons (2001), p.210, and also by Sharpe (1994), p.51.

³Similarly, Baillie and Kiliç (2006) use a logistic STR with the forward premium, money supply differentials, and conditional variance of US money growth as transition variables and find that a regime consistent with UIP exists. In the remainder of the paper, we mainly refer to Sarno et al. (2006), since their specification and interpretation is closer to our work.

carry-trades are a potential key driver for the surge in foreign exchange trading over the last years. Talking to practitioners and reading research and strategy reports published by financial institutions reveals that, in practice, capital is allocated to bias-trading strategies.⁴ While financial institutions usually do not fully disclose their approaches, the fact that bias-exploiting strategies have been used for years and that their use continues and even gets more widespread suggests (a certain degree of) economic significance. As a consequence, reconciliation of the findings related to the concept of limits to speculation with market evidence for bias-trading activity does not seem to be straightforward. One reason for this discrepancy might be that most academic research related to the forward bias is more concerned with modeling the puzzle rather than exploring how deviations from UIP can be exploited. Market participants, however, are certainly more interested in capitalizing on the bias and do not necessarily aim at finding an explanation for the anomaly (which they might simply take as given). Consequently, the approaches applied in practice might differ from those considered within the models investigated in the literature, leading to the diverse perceptions of whether trading the bias is attractive or not. In this paper we develop trading strategies explicitly aimed at exploiting the forward bias to investigate whether they allow for economically significant excess returns and thus should have the potential to attract capital.

3 Trading Strategies

In this Section we develop implementable bias-trading strategies used to assess the concept of limits to speculation. Our first approach is from the perspective of a (levered) proprietary trader facing the choice to either allocate funds to an optimal carry-trade portfolio or another investment opportunity. Our second approach considers active fund managers aiming at outperforming their benchmarks using currency option strategies to trade the bias. Both approaches, as well as modifications or combinations thereof, could be applied by other market participants (e.g. hedge funds or commodity trading advisors) as well.

3.1 Optimal Carry-Trade Portfolios

The forward bias suggests that high interest rate currencies tend to not depreciate as much as predicted by UIP and often even appreciate against lower interest rate currencies. A simple bias-exploiting strategy is the carry-trade, taking a long position in a deposit in the currency with the high interest rate, financed by the lower interest rate currency.⁵ In Appendix A we show that analogue implications

⁴We rely on conversations with practitioners and reports from Bank of America, Brevan Howard, Citigroup, Deutsche Bank, and Lehman Brothers. Further references outlining that the bias is viewed as exploitable by traders are given in e.g. Chinn (2006), p.10.

⁵Equivalently, one can sell forward the foreign currency if the forward premium is positive and vice versa.

for bias trading arise from nonlinear STR-models applied in recent research. The excess return of such a carry-trade (CT) can be written as

$$CT_{t+k} = \begin{cases} (1 + i_{t,k}^*)S_{t+k}/S_t - (1 + i_{t,k}) & \text{if } i_{t,k} < i_{t,k}^*, \\ (1 + i_{t,k}) - (1 + i_{t,k}^*)S_{t+k}/S_t & \text{if } i_{t,k} > i_{t,k}^*, \end{cases} \quad (3)$$

where $i_{t,k}$ and $i_{t,k}^*$ denote the effective k -period deposit rates available in the domestic and foreign money market.

Since diversification across multiple foreign currencies presumably improves performance as compared to trading limited to a single foreign currency, we consider a portfolio of bilateral carry-trades. Although the net investment of such a portfolio is zero, the portfolio is not implementable as a zero-investment due to margin requirements, i.e. some fraction of the notional value has to be held in a margin account that earns the domestic risk-free interest rate. For simplicity we assume that regulations require the full notional value to be held in a margin account.⁶ The return of a carry-trade portfolio (CTP) is thus given by the weighted average of the individual bilateral carry-trade excess returns plus the interest earned on the margin account, i.e. $CTP_{t+k} = \mathbf{w}'_t \mathbf{CT}_{t+k} + i_{t,k}$, where $\mathbf{w}'_t$ is a vector of portfolio weights representing the proportion of (overall) notional value allocated to the individual bilateral carry-trades and $\mathbf{CT}_{t+k}$ is a vector containing bilateral carry-trade excess returns as defined in Eq. (3).

Lyons (2001) reports that financial institutions often consider Sharpe ratios to choose between trading strategies and to measure their performance. Thus, our starting-point is the conventional mean-variance approach where the portfolio's Sharpe ratio is maximized. According to our discussions with practitioners, the decision to implement active trading strategies is, however, not exclusively based on the assessment of Sharpe ratios but also on covariance and downside risk.⁷ The latter is also relevant in the context of capital charges faced by financial institutions for market risks in their trading books.⁸ Hence, we constrain the portfolio to have a Value-at-Risk (VaR) not worse than that of the investment opportunity it is compared to. We use the historical simulation method which samples from historical data to calculate the VaR; see e.g. Jorion (2001).

A further restriction motivated by practical application is that the optimization is constrained to allocate weights across currencies in 5% units, since traders typically distribute funds in even proportions.

With the expected excess returns as specified below, the optimization problem is to find the portfolio

⁶Alternatively, this assumption can be viewed as requiring any fraction of the notional value as margin requirement and the remainder being invested at the risk-free interest rate.

⁷Here again, we rely on conversations with practitioners and reports from Bank of America, Brevan Howard, Citigroup, Deutsche Bank, and Lehman Brothers.

⁸For example, as stated in the *Amendment to the Capital Accord to Incorporate Market Risks* released by the Basel Committee on Banking Supervision (1996), banks fulfilling certain criteria are allowed to use internal models to measure their market risks for the purpose of calculating their capital charges. Banks have flexibility in devising the precise nature of their models but computing the Value-at-Risk is mandatory.

weights that maximize the portfolio's Sharpe ratio subject to the above constraints:

$$\begin{aligned}
& \max_{\mathbf{w}'_t} \frac{\mathbf{w}'_t \mathbf{R}_{t+k}^e}{\sqrt{\mathbf{w}'_t \boldsymbol{\Sigma}_t \mathbf{w}_t}} \\
& \text{subject to} \\
& VaR_t(\alpha) \geq VaR_t^{Benchmark}(\alpha), \\
& w_t \in \{0, 0.05, 0.1, \dots, 0.95, 1\}, \quad \text{and} \\
& \mathbf{w}'_t \mathbf{1} = 1,
\end{aligned} \tag{4}$$

where $\mathbf{R}_{t+k}^e$ denotes a vector of expected excess returns, $\boldsymbol{\Sigma}_t$ the associated variance-covariance matrix, and $VaR_t(\alpha)$ and $VaR_t^{Benchmark}(\alpha)$ denote the VaRs of the optimized portfolio and the investment opportunity the portfolio is compared to.⁹

For generating their expected excess returns, carry-traders presume that the exchange rate follows a random walk, i.e. an appreciation has the same probability as a depreciation implying that the expected change is zero. In this case, the expected excess return resulting from Eq. (3) corresponds to the interest rate differential implying that it is rational to finance a long deposit in the high interest rate currency by a short position in the lower interest rate currency.

Furthermore, we consider a second strategy which does not persistently bet against UIP - but rather on the reversion of temporary deviations from a long-run UIP equilibrium - by relying on spot rate forecasts from vector error correction models (VECMs). These type of models have been extensively used in past research related to exchange rate behavior and appear to capture the empirical stylized facts reasonably well; see e.g. Brenner and Kroner (1995) and Zivot (2000). The application of VECMs is based on the idea that UIP is a long-run equilibrium disturbed by temporary (short-run) exogenous shocks. We use recursive VECMs to generate spot rate forecasts which are used to calculate the expected return from an uncovered foreign investment. The specification of the error correction model can be found in Appendix B. If the expected return is higher than the domestic deposit rate, a long foreign deposit is financed by a short domestic deposit and vice versa.

To solve the optimization problem we use a heuristic optimization algorithm called Threshold Accepting, since conventional optimization approaches are not capable of handling the imposed constraints. Threshold Accepting, introduced by Dueck and Scheuer (1990), is a refined local search algorithm, which enables to solve portfolio optimization problems subject to nearly arbitrary constraints and almost every utility function. The algorithm is able to escape local minima by accepting solutions which undershoot the current solution by not more than a given threshold. During the course of the algorithm the threshold is successively reduced, reaching the value of zero eventually. The algorithm

⁹The results reported in our paper rely on recursively calculated (sample) standard deviations. We have also calculated volatilities based on exponentially weighted moving averages which leads to results exhibiting minor quantitative differences without changing conclusions

yields an optimal solution or a solution sufficiently close to an optimal one; convergence is assured as shown by Althoefer and Koschnick (1991).¹⁰ An outline of our heuristic portfolio optimization algorithm can be found in Appendix C.

3.2 Options Strategy

The application of the Black and Scholes (1973) framework to exchange rates by Garman and Kohlhagen (1983) yields the following prices for European call (C) and put (P) options:

$$\begin{aligned} C_t &= F_{t,k} \exp[-r_t k] \Phi(d_1) - X \exp[-r_t k] \Phi(d_2) \quad \text{and} \\ P_t &= X \exp[-r_t k] \Phi(-d_2) - F_{t,k} \exp[-r_t k] \Phi(-d_1), \end{aligned} \tag{5}$$

with

$$d_1 = \frac{\ln(F_{t,k}/X) + 0.5\sigma^2 k}{\sigma\sqrt{k}} \quad \text{and} \quad d_2 = d_1 - \sigma\sqrt{k},$$

where r_t denotes the continuously compounded domestic risk-free interest rate, Φ the cumulative standard-normal distribution function, X the strike price, and σ the volatility of the spot rate.

Since the forward rate serves as the risk-adjusted expectation of the future spot rate within the option pricing framework in Eq. (5), the finding that the forward rate is a biased predictor of the future spot rate is relevant for currency option pricing and corresponding trading strategies. The forward bias implies that the forward rate overestimates (underestimates) the future spot rate if the domestic interest rate is higher (lower) than the foreign. Considering the partial derivatives of call and put prices with respect to the forward rate, $\frac{\partial C_t}{\partial F_{t,k}} \geq 0$ and $\frac{\partial P_t}{\partial F_{t,k}} \leq 0$, indicates that calls would be overpriced and puts underpriced if the domestic interest rate is higher than the foreign and vice versa.

Consider now a call and a put on the same currency pair with the same maturity and the strike prices for both set equal to the forward rate. According to the option pricing framework in Eq. (5) both prices are the same and given by

$$\begin{aligned} C_t &= P_t = F_{t,k} \exp[-r_t k] [\Phi(d_1) - \Phi(d_2)], \\ \text{with} \quad d_1 &= 0.5\sigma\sqrt{k} \quad \text{and} \quad d_2 = -d_1. \end{aligned} \tag{6}$$

Given the above stated implications of the forward bias, however, we would not ascribe the same probability to both options to end up in-the-money and hence also not the same prices. Considering the case that the domestic interest rate is higher than the foreign, we would presume that the put is

¹⁰For a comprehensive introduction to Threshold Accepting see Winker (2001); the application to portfolio optimization has been introduced by Dueck and Winker (1992).

more likely to be in-the-money at maturity than the call even though both have the same price. In fact, given an estimate of the forward bias, with $bias_{t,k} \equiv S_{t+k} - F_{t,k}$, we would expect that only a call with a strike price lower than $F_{t,k} + \widehat{bias_{t,k}}$ ends up in-the-money at maturity. An attempt to exploit the bias is therefore to go long a put with strike price equal to the forward rate and, at the same time, go short a call with strike price equal to $F_{t,k} + \widehat{bias_{t,k}}$. To limit the downside risk, a partly offsetting long position in a forward contract is added at the cost of giving up some upside potential. The premium collected for selling the call exceeds the premium paid for buying the put, since $\widehat{bias_{t,k}} < 0$ and $\frac{\partial C_t}{\partial X} \leq 0$; the forward contract has an initial value of zero. Thus, at initiation of the strategy the difference between the option premia is collected and, at maturity, the outflow ranges from the difference in the strike prices $|\widehat{bias_{t,k}}|$ in the worst case (if $bias_{t,k} \geq 0$) to zero in the best case (if $bias_{t,k} \leq \widehat{bias_{t,k}}$).¹¹ The payoff of this strategy resembles that of a bear spread.¹² Analogue arguments apply if the foreign interest rate is higher than the domestic, resulting in a strategy payoff resembling that of a bull spread.

The success of this strategy crucially depends on the ability to predict the forward bias. Thus, inspired by the argumentation and payoff schemes from above, but avoiding the necessity of an estimated magnitude of the bias, we consider the use of digital options. Relying solely on the sign of the interest rate differential as signal, i.e. the estimated sign of the bias, we use digital calls (DC) and puts (DP) instead of bull and bear spreads with payoffs and prices given by

$$DC_{t+k} = \begin{cases} 1 & \text{if } S_{t+k} > X = F_{t,k} \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad DC_t = \exp[-r_t k] \Phi(d_2) \quad \text{and} \quad (7)$$

$$DP_{t+k} = \begin{cases} 1 & \text{if } S_{t+k} < X = F_{t,k} \\ 0 & \text{otherwise} \end{cases} \quad \text{and} \quad DP_t = \exp[-r_t k] \Phi(-d_2). \quad (8)$$

We consider a strategy that finances a long position in a digital call with strike price equal to the forward rate if the foreign interest rate is higher than the domestic and in a digital put otherwise. The excess returns of this zero-investment options strategy (OS) can be written as

$$OS_{t+k} = \begin{cases} DC_{t+k}/DC_t - (1 + i_{t,k}) & \text{if } i_{t,k} < i_{t,k}^*, \\ DP_{t+k}/DP_t - (1 + i_{t,k}) & \text{if } i_{t,k} > i_{t,k}^*. \end{cases} \quad (9)$$

However, pure options strategies result in extreme return distributions, i.e. the potential of high

¹¹The strategy can also be constructed such that there is an initial outflow and an inflow at maturity.

¹²Comparing such a bear spread to a standard put with strike price equal to the forward rate, shows that the payoff is the same for both if $bias_{t,k} \geq \widehat{bias_{t,k}}$. For $bias_{t,k} < \widehat{bias_{t,k}}$, however, the payoff from the bear spread is capped by the difference of the strike prices $|\widehat{bias_{t,k}}|$. Of course, the additional upside potential of the put comes at the cost of a higher premium. If the estimated bias is adequate, it is unlikely that this additional initial cost will be compensated by higher payoffs.

returns is accompanied by great ‘risk’. Although the expected return for taking this risk might be adequate, one might not consider them as interesting (due to the level of risk itself). Nevertheless, they might be promising for active fund managers trying to outperform their benchmarks. Hence, we consider fund managers investing a fraction, λ , of their funds in the options strategy and the remainder, $1 - \lambda$, in a benchmark-tracking portfolio. For such an options strategy enhanced portfolio (*OSEP*) the return can be written as

$$OSEP_{t+k} = (1 - \lambda) R_{t,k}^{Benchmark} + \lambda \begin{cases} DC_{t+k}/DC_t - 1 & \text{if } i_{t,k} < i_{t,k}^*, \\ DP_{t+k}/DP_t - 1 & \text{if } i_{t,k} > i_{t,k}^*. \end{cases} \quad (10)$$

where $R_{t,k}^{Benchmark}$ denotes the return of the benchmark-tracking portfolio. The allocation of funds between the options strategy and the benchmark-tracking portfolio depends on the fund manager’s risk tolerance and confidence in the options strategy.

4 Empirical Analysis

Our empirical analysis is based on monthly observations of spot and 1-month forward USD exchange rates against the Australian Dollar (AUD), Canadian Dollar (CAD), Swiss Franc (CHF), Danish Krone (DKK), British Pound (GBP), Japanese Yen (JPY), and Euro (EUR). Furthermore, we use 1-month deposit rates and 1-month implied option volatilities.¹³ Moreover, we take monthly observations of the S&P 500 Index (S&P500), the MSCI World Total Return Index (MSCI), and the JP Morgan US Government Bond Index with maturities from 7 to 10 years (JPMGB), all denominated in USD. All our data are end of month bid-ask data collected from traders by Reuters.¹⁴ Due to data availability, our evaluation period spans from 01/1990 to 12/2005 for the optimal carry-trade portfolios and from 01/1995 to 12/2005 for the options strategies.¹⁵ In order to check for the robustness of our findings, we consider subsamples.

All our results are out-of-sample in the sense that trading decisions are only based on information available at that time. Furthermore, at the beginning of the evaluation period more than a decade

¹³We rely on over-the-counter rather than on exchange-traded options data, since liquidity is substantially higher in the over-the-counter market; see e.g. Bank for International Settlements (2002) and Jeanneau (2003). Due to data availability, we use implied volatilities from standard options instead of digital options. To account for potentially larger transaction costs for trading digital options, we also include bid-ask spreads in interest rates when calculating option prices. Looking at recent digital options data indicates that this approach is rather conservative.

¹⁴Papers related to ours often use Datastream data for their analysis. However, we explicitly wanted to incorporate transaction costs (bid-ask quotes) which are only available in Reuters for all time series considered in our paper. Comparing the mid rates of the Reuters data to those from Datastream shows that there are almost no differences.

¹⁵Exchange rate and interest rate time series are available starting from the early to mid 1980s. Since we require a minimum of five years of data for generating the first window for the recursive calculation of standard deviations and VaRs, our evaluation period for the optimal currency portfolios starts with 01/1990. Furthermore, we use the European Currency Unit (ECU) for the time period prior to the introduction of the EUR to allow for continuity in the calculations related to VaRs and VECMs. The German Mark instead of the ECU yields similar results. For the EUR the options strategies evaluation period spans from 01/1999 to 12/2005.

has already been spent on research related to the forward bias. Hence, traders have back then already been well aware of the signals that we use for the trading strategies.

Our data exhibits the forward bias as documented in the literature. The estimates of the Fama regression Eq. (2) show that the slope coefficient is always negative and significantly different from unity, indicating not only that UIP does not hold but also that the higher interest rate currency tends to appreciate rather than depreciate. Across the currencies investigated, annualized Sharpe ratios of bilateral carry-trades, as described in Eq. (3), are 0.43 on average and therefore below the practitioner threshold of 0.5 quoted by Lyons (2001). Thus, one cannot conclude that trading the forward bias on a bilateral basis is attractive in general, being consistent with the concept of limits to speculation.

4.1 Optimal Carry-Trade Portfolios

In Table 1 we report the annualized results of the optimal carry-trade portfolios, as described in Eq. (4), with the downside risk being constrained to be not worse than that of the respective benchmark, i.e. the S&P500, the MSCI, or the JPMGB.¹⁶ Table 1 also reports the results of the portfolios with expected returns derived from VECM spot rate forecasts. To investigate whether the optimal portfolios outperform the respective benchmark, we use the tests proposed by Jobson and Korkie (1981) with the correction pointed out by Memmel (2003), denoted JKM, and Gibbons et al. (1989), denoted GRS.

[Insert Table 1 here]

The full evaluation period Sharpe ratios for the benchmarks range from 0.11 for the MSCI, to 0.32 for the S&P500, up to 0.54 for the JPMGB. The corresponding VECM-based portfolio Sharpe ratios are 0.71, 0.71, and 0.62 - hence in all cases above their respective benchmark, although not always significantly. The carry-trade portfolios yield Sharpe ratios of 0.97, 0.97, and 0.93, which are in all cases significantly above their benchmarks. These results are mainly driven by the lower volatility of the optimal portfolios, in particular when compared to the equity indices. In terms of VaR, both strategies are slightly better than the bond index and substantially better than the equity indices. Finally, we find that CAPM-style β s (with respect to the benchmark used in the optimization) are not significantly different from zero, indicating that the risk of these strategies is largely diversifiable.

The subsample results suggest that these findings are robust over time. The optimal portfolios are generally characterized by higher Sharpe ratios as well as lower VaRs than the benchmarks, and do not exhibit systematic risk.¹⁷ To visualize the robustness of our results, we plot the cumulative risk-adjusted excess returns for the carry-trade portfolio with the S&P500 and the JPMGB as benchmarks;

¹⁶We quantify downside risk as the 5% historical VaR, calculated on a monthly basis with a recursively extending time window.

¹⁷The JKM and GRS test generally support each other. Nevertheless, in some cases the JKM statistic doubts GRS-significance which can be attributed to small sample problems associated with the JKM test; see Memmel (2003).

see Figure 1. While the performance of the carry-trade portfolios is less pronounced in the first years of our sample, we find a rather steady accumulation of risk-adjusted excess returns from 1995 onwards.

[Insert Figure 1 here]

As mentioned at the beginning of our empirical analysis, bilateral carry-trades do not necessarily yield economically significant excess returns. Our optimal carry-trade portfolios, however, persistently outperform their benchmarks in terms of Sharpe ratios, have lower downside risk, and do not exhibit systematic risk. While these findings suggest that traders should show interest in such strategies, which is consistent with market evidence that carry-trading is a potential key driver for the surge in foreign exchange trading, they challenge the concept of limits to speculation.

Sarno et al. (2006) test some of the general predictions of the limits to speculation hypothesis in a STR framework. In their setup, large (expected) deviations from UIP are followed by reversions towards UIP, whereas small deviations lead to a statistically persistent but economically unimportant forward bias. Empirically they find that UIP does not hold most of the time which corresponds to the latter regime. According to their calculations, based on an average across five USD exchange rates, a shift towards the UIP regime is induced only if the annual Sharpe ratio exceeds 1.69 (which corresponds to the - relatively low - transition function value of 0.5). Our strategies yield Sharpe ratios close to 1 and would therefore be consistent with a persistent bias.¹⁸ The performance of our strategies is indeed persistent but at the same time also relevant in economic terms since other investments are outperformed significantly.¹⁹

Related UIP research often tries to attribute the forward bias and related excess returns to a (time-varying) risk premium.²⁰ Attempts of modeling such risk-premia, however, usually imply too high degrees of risk aversion; see e.g. the surveys of Hodrick (1987), Froot and Thaler (1990), Lewis (1995), Engel (1996), and Sarno (2005). Since the concept of limits to speculation suggests that traders compare the performance of bias-trading strategies to that of other investment opportunities, one would expect to find a similar degree of risk aversion across markets in which traders are active. Presuming that traders are mean-variance optimizers - as we do in our analysis - one can reverse engineer the objective function to obtain the implied coefficient of relative risk aversion for a given portfolio as the average excess return divided by its variance. We report this measure in the last column of Table 1 and find that the relative risk aversion implied by the currency strategies is substantially

¹⁸Sarno et al. (2006) use a somewhat different data set. However, our subsequent line of reasoning is rather general and therefore independent of these differences.

¹⁹A reason for this discrepancy could be that - as Sarno et al. (2006) acknowledge - the STR-type nonlinearity might not be exclusively rooted in limits to speculation but might also stem from other sources such as transaction costs or central bank intervention.

²⁰If investors were risk-averse, the Fama-regression in Eq. (2) would suffer from an omitted variable bias. The relevant covariance due to which the risk premium arises is between exchange rates and the market portfolio in asset market models, between exchange rates and growth in aggregate consumption in real economy models, and across exchange rates in portfolio balance models.

higher than that implied by the benchmarks. While this finding seems to mirror the problems arising in attempts to explain the forward bias by risk-premia, it is of little support for the concept of limits to speculation which suggests that the same traders are active across markets.

4.2 Options Strategy

In Table 2 we report monthly results of the bilateral strategies based on digital options as described in Eq. (9). Additionally, we consider an equally-weighted (i.e. naive) portfolio of these strategies to gain some insight regarding possible benefits from diversification across currencies.²¹

[Insert Table 2 here]

The full evaluation period average excess returns for the bilateral strategies range from 9.69% for the CHF to 23.48% for the AUD per month; for the shorter evaluation period in case of the EUR the average excess return is 34.56%. Since average positive and negative excess returns are around +100% and -100% respectively, the performance of the strategies is driven by their hit-ratio, i.e. by the fraction of cases in which positive excess returns are generated. Across currencies we find hit-ratios between 55% and 62%, in case of the EUR the hit-ratio is 67%. As an indicator for the significance of the hit-ratios we simulate coin-flipping strategies, i.e. at each point in time the decision which position to take is not based on the interest differential but on flipping a coin. For each currency we simulate 10,000 coin-flipping time series (each with the same length as the actual sample) and calculate the empirical probabilities of achieving the actually realized or a better hit-ratio. The probabilities vary between 0.001 and 0.091, clearly indicating that the profits of our strategies are not a result of pure chance.

For the naive portfolio, we find an average excess return of 18.48% which results on the one hand from the average positive excess return being about 44% while the average negative excess return is only around -34%, and on the other from a significant hit-ratio of 67%. In Figure 2 we plot the probability densities of excess returns for the naive portfolio. While excess returns of the bilateral strategies yield bimodal distributions with just two bins at around -100% and +100%, with the mode on the positive side being higher than that on the negative, the portfolio excess returns are normally distributed as judged by the Jarque-Bera test.

[Insert Figure 2 here]

²¹Since the distribution of excess returns of digital options is bimodal, standard deviations and hence Sharpe ratios are not very meaningful, thus we consider other statistics characterizing the distribution of excess returns. Furthermore, we report results on a monthly basis to highlight distributional features.

The subsample results indicate that the aforementioned findings are robust over time. For the bilateral strategies, average excess returns exhibit variability for some of the currencies, while for others they are highly persistent. Out of 20 cases we only find 4 in which the hit-ratio is 50% or lower (47% in the worst case), leading to negative average excess returns over the subsample. The results of the naive portfolio are persistent across subsamples. Average excess returns range from about 15% to 21%, resulting from average positive excess returns being always higher than average negative excess returns (in absolute terms) and from significant hit-ratios of 64% to 73%.

While a pure options strategy might be distracting due to the level of risk itself (even if the return for taking this risk might be adequate), the strategy might be a promising supplement for active fund managers seeking to outperform their benchmarks. In Table 3 we report annualized results for allocating 1% of funds to the options strategies and the remaining 99% in a portfolio tracking the S&P500, the MSCI, or the JPMGB. To investigate whether the enhanced portfolios outperform the respective benchmark, we use the JKM and GRS tests as well as Jensen's alpha.

[Insert Table 3 here]

The full evaluation period Sharpe ratios of the enhanced portfolios are generally significantly higher than those of their respective benchmarks. The improvement is mainly driven by an increase in the return whereas the volatility remains virtually unchanged. Jensen's alpha is positive in all cases, with varying estimates and varying degrees of significance across currencies, though. The enhancement with the naive portfolio yields for all benchmarks very similar results: an additional return of around 2.2% per year without increasing volatility and a highly significant Jensen's alpha of about 0.0019 per month.

The results for the subsamples suggest, consistent with the pure options strategies from above, that these findings are robust over time. As an example, we plot the cumulative risk-adjusted excess returns for the S&P500 investment enhanced with the naive portfolio in Figure 3.

[Insert Figure 3 here]

The results of our options strategy are also interesting in the context of previous research related to limits to speculation. For instance, Sarno et al. (2006) argue that a statistically persistent but economically small forward bias exists most of the time. Our approach allows to exploit an arbitrarily small forward bias, since the performance solely relies on whether a forward bias exists but not on its magnitude; the associated risk is substantial, though. However, allocating just 1% of funds to the options strategies and the remainder in a benchmark-tracking portfolio, allows fund managers to significantly outperform their benchmarks at a virtually unchanged volatility level.

5 Conclusions

The concept of limits to speculation suggests that the forward bias might be significant statistically but nevertheless unimportant in economic terms. In particular, it is argued to be too small to attract speculative capital, since other investment opportunities yield higher Sharpe ratios. As a consequence, traders leave the bias unexploited and statistically persistent. Market evidence, on the contrary, suggests that the bias is traded in practice and that carry-trades might even be a key driver for the surge in foreign exchange trading over the last years.

In this paper we develop implementable bias-trading strategies in bilateral and multi-currency settings. We use a constrained portfolio optimization over carry-trades as well as approaches based on combinations of currency options. Our empirical analysis shows that the bias does not only exist statistically but that betting against UIP yields Sharpe ratios higher than those of reasonable benchmarks, even after adjusting for transaction costs. Bias-trading strategies can be employed as attractive investment strategies per se (higher Sharpe ratios), useful diversification devices (no systematic risk), and promising portfolio extensions for active fund managers trying to outperform their benchmarks (higher Sharpe ratios and positive Jensen's alpha).

Overall, our findings mirror the problems arising in attempts to explain the forward bias by risk-premia and are consistent with market evidence that the bias is traded. As a consequence, the limits to speculation hypothesis seems - despite its intuitive appeal - unlikely to provide a (stand-alone) explanation for the persistence of the forward bias.

Appendix

A Implications of STR-Type Nonlinearity for Bias-Trading

Given the evidence in favor of STR-type nonlinearity in the relation between spot and forward rates, see e.g. Baillie and Kiliç (2006) and Sarno et al. (2006), we investigate whether the trading implications from a STR framework are similar to those from the Fama regression.

The exponential STR framework specified in Sarno et al. (2006) can be re-parameterized in terms of excess returns over UIP:

$$ER_{t+1} = [\alpha_1 + (\beta_1 - 1)(f_{t,1} - s_t)] + [\alpha_2 + \beta_2(f_{t,1} - s_t)] F(ER_{t+1}^e) + v_{t+1}, \quad (11)$$

where v_{t+1} denotes a disturbance term. The transition function $F(\cdot)$, with the expected excess return, ER_{t+1}^e , as the transition variable, is given by $F = 1 - \exp[-\gamma(ER_{t+1}^e)^2]$, with $\gamma > 0$ determining the smoothness of the transition between regimes. $F(\cdot)$ is symmetrically inverse-bell shaped around zero with $F(0) = 0$ and $\lim_{x \rightarrow \pm\infty} F(x) = 1$. Hence, the transition function enables a smooth transition between a ‘lower regime’, $F(\cdot) = 0$, and an ‘upper regime’, $F(\cdot) = 1$. Combining the above equation with the restrictions $\alpha_2 = -\alpha_1$ and $\beta_2 = 1 - \beta_1$ and rearranging yields²²

$$ER_{t+1} = \alpha_1 [1 - F(ER_{t+1}^e)] + \underbrace{(\beta_1 - 1)(f_{t,1} - s_t)}_{\beta^*} + \underbrace{(1 - \beta_1) F(ER_{t+1}^e)(f_{t,1} - s_t)}_{\beta^{**}} + v'_{t+1}, \quad (12)$$

where v'_{t+1} denotes a disturbance term.²³ To derive trading signals corresponding to those based on the Fama regression, we consider the forward premium, α_1 , β^* , β^{**} , and the value of the transition function. Given that α_1 is typically very close to zero and recalling that $F(\cdot)$ is bounded between 0 and 1, the following is true:

1. β^* is only equal to β^{**} if $\beta_1 = 1$, in which case $\beta^* = \beta^{**} = 0$;
2. If $\beta_1 \neq 1$, β^{**} always has the opposite sign of β^* , with $\beta^* < 0$ for $\beta_1 < 1$;
3. $|\beta^*| \geq |\beta^{**}|$, in particular $\beta^* = -\beta^{**}$ if $F(\cdot) = 1$ and / or $\beta_1 = 1$ and $|\beta^*| > |\beta^{**}|$ if $F(\cdot) < 1$ and $\beta_1 \neq 1$.

As a consequence the sign of the excess return is determined by the sign of the forward premium, the sign of $\beta^* = \beta_1 - 1$, and the value of the transition function. The empirical results of Sarno et al.

²²Under $\alpha_2 = -\alpha_1$ and $\beta_2 = 1 - \beta_1$, as considered by Baillie and Kiliç (2006) and Sarno et al. (2006), the lower regime corresponds to a domain where the STR reduces to the standard linear Fama regression, whereas in the upper regime the STR becomes a different linear Fama regression with parameters consistent with UIP.

²³The STR framework is consistent with the evidence that spot and forward rates are cointegrated, since within this framework larger expected deviations from UIP would attract capital thereby inducing reversion towards UIP - the long-run equilibrium in a corresponding cointegration system.

(2006) show that β_1 is typically negative or at least less than unity along with a value of the transition function less than unity most of the time. Thus, if the true data generating process governing the relation between spot and forward rates were of the nonlinear form as in the STR framework, one would sell forward the foreign currency if the forward premium is positive and vice versa. Hence, the implications for bias-trading are the same for the linear and nonlinear model.²⁴

B Vector Error Correction Model

Forward contracts are usually priced using covered interest rate parity, which, as every linear no-arbitrage pricing formula, combines assets to duplicate another, redundant asset. The redundant asset should therefore have the same stochastic trend as the original asset (if any), as pointed out by Brenner and Kroner (1995). As shown by Engle and Granger (1987), the presence of common stochastic trends, i.e. cointegration, requires the employment of error correction models. Although the argumentation of Brenner and Kroner (1995) implies cointegration at any lead or lag of the spot and forward rate, Zivot (2000) argues that models of cointegration between s_t and $f_{t,k}$ are better suited to capture the empirical stylized facts of exchange rate behavior as compared to models of cointegration between s_{t+k} and $f_{t,k}$. We, therefore, use recursive models of cointegration between s_t and $f_{t,k}$. If s_t and $f_{t,k}$ are not cointegrated, the interest rate differential is included in the system, as suggested by Brenner and Kroner (1995). If there is still no cointegration relation, a vector autoregressive model (VARM) is used. We apply the conventional Johansen (1995) procedure to determine the particular type of model to be employed.

Our analysis is based on the following standard VECM specification:

$$\Delta \mathbf{y}_t = \alpha(\beta \mathbf{y}_{t-1} + \mu + \rho t) + \sum_{i=1}^{p-1} \Gamma_i \Delta \mathbf{y}_{t-i} + \gamma + \tau t + \mathbf{e}_t, \quad (13)$$

where $\Delta \mathbf{y}_t$ denotes a $K \times 1$ vector of potentially cointegrated variables, α and β denote $K \times r$ matrices of rank r , with r being the number of linearly independent cointegrating vectors, μ and ρ denote $r \times 1$ vectors of parameters, p denotes the lag order of the underlying VARM, $\Gamma_i = -\sum_{j=i+1}^p \mathbf{A}_j$, with $\mathbf{A}_j$ being $K \times K$ matrices of parameters of the underlying VARM, γ and τ denote $K \times 1$ vectors of parameters, and $\mathbf{e}_t$ denotes a $K \times 1$ vector of disturbances.

To select the lag order of the underlying VARM we use Lütkepohl's version of Schwarz's Bayesian information criterion; see Lütkepohl (1993). Given the lag order, we use Johansen's trace statistic method to determine the number of linearly independent cointegrating vectors. If $0 < r < K$, the variables in $\mathbf{y}_t$ are cointegrated.²⁵ If $r = 0$, a VARM in the first differences is adequate, and if $r = K$,

²⁴Note that our conclusions are independent of the particular type of transition function or transition variable.

²⁵Placing restrictions on the constant and trend terms in the above VECM results in the following five cases, see Johansen (1995): (i) unrestricted trend, (ii) restricted trend, $\tau = 0$, (iii) unrestricted constant, $\tau = 0$ and $\rho = 0$, (iv)

a VAR in the levels is used.

C Threshold Accepting Algorithm

Dueck and Scheuer (1990) introduced Threshold Accepting and the first paper to apply Threshold Accepting to portfolio optimization was Dueck and Winker (1992). For recent research in this direction see e.g. Gilli and Kellezi (2002). In the context of the optimization problem in (4), the Threshold Accepting algorithm is designed as follows. After choosing a random starting portfolio, the neighborhood of this portfolio is explored. A random change in the portfolio composition is performed such that, if feasible, the weight of one currency will be reduced by 5% and the weight of another currency will be increased by 5%. The resulting new portfolio will be accepted as solution, if the evaluation of the new composition does not undershoot the evaluation of the old composition by more than a given threshold. The evaluation is based on the Sharpe ratio but includes a penalty function such that portfolios violating the VaR constraint are not accepted. This procedure is repeated for a fixed number of steps during a fixed number of iterations. For subsequent iterations the threshold is successively reduced, reaching the value of zero eventually. By accepting solutions which undershoot the current solution by not more than a given threshold, the algorithm is able to escape local minima. With $f(\cdot)$ denoting the objective function, the pseudo-code of the Threshold Accepting algorithm is given in Algorithm 1.

Algorithm 1: Threshold Accepting

Pseudo-Code for Threshold Accepting Algorithm
. Fix a number of iterations $nriter$ and steps $nrsteps$
. Fix a sequence of thresholds $T_i, i = 1, \dots, nriter$
. Choose randomly a starting portfolio $\mathbf{w}^0$
. For $i = 1$ to $nriter$
. For $s = 1$ to $nrsteps$
. Choose randomly $\mathbf{w}^1$ from the neighborhood of $\mathbf{w}^0$
. If $f(\mathbf{w}^1) < f(\mathbf{w}^0) + T_i$, then
. $\mathbf{w}^0 = \mathbf{w}^1$
. End if
. End for
. End for

The algorithm yields an optimal solution or a solution sufficiently close to an optimal one; convergence is assured as shown by Althoefer and Koschnick (1991). For a comprehensive introduction to Threshold Accepting see Winker (2001).

restricted constant $\tau = 0$, $\rho = 0$, and $\gamma = 0$, and (v) no trend $\tau = 0$, $\rho = 0$, $\gamma = 0$, and $\mu = 0$. Since a quadratic trend in the levels seems to be economically unreasonable, we a priori exclude the unrestricted trend case. Thus, we determine the number of linearly independent cointegrating vectors for each of the remaining four cases. Given that the variables in $\mathbf{y}_t$ are cointegrated, i.e. $0 < r < K$, we use Schwarz's Bayesian information criterion to determine the most plausible trend assumption.

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Table 1: Optimal Carry-Trade Portfolios

Benchmark	$\bar{R}$	SD	SR	VaR	$p[JKM]$	$p[GRS]$	β_{CAPM}	SE_{β}	$Imp.RA$
Panel A. Full Evaluation Period (01/1990 - 12/2005)									
S&P500	8.93%	14.27%	0.32	-6.78%					2.24
CTP	9.44%	5.25%	0.97	-2.06%	0.0042	0.0000	-0.0306	0.0264	18.48
VECM	8.55%	5.92%	0.71	-2.52%	0.0877	0.0000	-0.0470	0.0299	11.99
MSCI	6.00%	14.23%	0.11	-6.28%					0.77
CTP	9.43%	5.25%	0.97	-2.06%	0.0003	0.0000	-0.0296	0.0264	18.48
VECM	8.56%	5.92%	0.71	-2.52%	0.0152	0.0000	-0.0285	0.0300	11.99
JPMGB	7.73%	6.21%	0.54	-2.76%					8.70
CTP	9.09%	5.11%	0.93	-1.87%	0.0000	0.0000	0.0816	0.0592	18.20
VECM	7.89%	5.67%	0.62	-2.17%	0.1945	0.0672	0.1110	0.0657	10.93
Panel B. Subsample 1 (01/1990 - 12/1995)									
S&P500	9.98%	11.80%	0.41	-7.67%					3.47
CTP	8.68%	5.74%	0.62	-1.92%	0.3532	0.1031	-0.0304	0.0570	10.80
VECM	6.36%	6.06%	0.20	-2.30%	0.6310	(0.3482)	-0.0467	0.0603	3.30
MSCI	5.28%	14.05%	0.01	-6.66%					0.07
CTP	8.68%	5.74%	0.62	-1.92%	0.1640	0.0022	0.0084	0.0477	10.80
VECM	6.36%	6.06%	0.20	-2.30%	0.3827	0.9103	0.0191	0.0506	3.30
JPMGB	9.94%	5.99%	0.80	-2.84%					13.36
CTP	8.07%	5.64%	0.53	-1.63%	0.8884	(0.0165)	0.1500	0.1093	9.40
VECM	5.57%	5.85%	0.07	-1.85%	0.9994	(0.0000)	0.1839	0.1131	1.20
Panel C. Subsample 2 (01/1996 - 12/2000)									
S&P500	16.63%	16.05%	0.69	-5.52%					4.30
CTP	10.56%	5.19%	0.96	-2.07%	0.3825	0.0355	-0.0371	0.0422	18.50
VECM	10.73%	5.79%	0.89	-2.52%	0.4171	0.1327	-0.0232	0.0473	15.37
MSCI	11.22%	14.20%	0.40	-5.78%					2.82
CTP	10.53%	5.19%	0.96	-2.07%	0.2729	0.0002	-0.1021	0.0460	18.50
VECM	10.75%	5.79%	0.90	-2.52%	0.2920	0.0009	-0.0514	0.0531	15.54
JPMGB	6.48%	5.56%	0.17	-2.72%					3.06
CTP	10.25%	5.08%	0.92	-1.95%	0.0067	0.0000	-0.1049	0.1204	18.11
VECM	10.68%	5.69%	0.90	-2.31%	0.0169	0.0001	-0.1253	0.1348	15.82
Panel D. Subsample 3 (01/2001 - 12/2005)									
S&P500	-0.01%	14.91%	-0.15	-6.98%					-1.01
CTP	9.22%	4.73%	1.47	-2.22%	0.0151	0.0000	-0.0189	0.0412	31.08
VECM	9.00%	5.89%	1.14	-2.78%	0.1052	0.0000	-0.0716	0.0510	19.35
MSCI	1.65%	14.58%	-0.04	-6.32%					-0.27
CTP	9.22%	4.73%	1.47	-2.22%	0.0153	0.0000	-0.0033	0.0422	31.08
VECM	9.00%	5.89%	1.14	-2.78%	0.1154	0.0000	-0.0618	0.0524	19.35
JPMGB	6.32%	7.09%	0.57	-2.70%					8.04
CTP	9.15%	4.50%	1.53	-2.08%	0.0001	0.0000	0.1400	0.0812	34.00
VECM	7.89%	5.42%	1.03	-2.41%	0.0508	0.0009	0.2067	0.0971	19.00

Notes: In each Panel we report, for the time period indicated in the header, the annualized results of the carry-trade portfolio (CTP) and the VECM-based portfolio (VECM) as compared to the S&P500, MSCI, and JPMGB. $\bar{R}$ denotes the average return, SD the corresponding standard deviation, SR the resulting Sharpe ratio, and VaR the 5% historical VaR calculated recursively. Furthermore, $p[JKM]$ states the significance level at which the strategy's Sharpe ratio is higher than that of the respective benchmark, using Jobson and Korkie (1981) and Memmel (2003). $p[GRS]$ is the significance level at which the strategy has a higher (lower) Sharpe ratio, using Gibbons et al. (1989). β_{CAPM} denotes a CAPM-style β (with respect to the benchmark used in the optimization) and SE_{β} the corresponding standard error. Finally, $Imp.RA$ is the implied coefficient of relative risk aversion obtained from reverse engineering the mean-variance objective function.

Table 2: Pure Options Strategies

Strategy	$\bar{R}$	$Hit-Ratio$	$p[Hit-Ratio]$	$\bar{R}$ if $R < 0$	$\bar{R}$ if $R > 0$
Panel A. Full Evaluation Period (01/1995 - 12/2005)					
AUD	23.48%	61.83%	0.002	-100.36%	99.92%
CAD	15.62%	58.02%	0.032	-100.35%	99.54%
CHF	9.69%	55.73%	0.066	-100.33%	97.11%
DKK	23.12%	61.83%	0.002	-100.37%	99.35%
GBP	11.99%	55.73%	0.066	-100.36%	101.26%
JPY	18.26%	54.96%	0.092	-100.33%	115.44%
EUR ^a	34.59%	67.47%	0.001	-100.27%	99.61%
Naive	18.48%	67.18%	0.001	-33.90%	44.08%
Panel B. Subsample 1 (01/1995 - 12/1998)					
AUD	26.73%	63.83%	0.001	-100.48%	98.81%
CAD	18.17%	59.57%	0.009	-100.47%	98.68%
CHF	4.66%	53.19%	0.214	-100.47%	97.17%
DKK	-7.32%	46.81%	0.733	-100.47%	98.52%
GBP	20.24%	59.57%	0.009	-100.47%	102.16%
JPY	25.93%	57.45%	0.032	-100.47%	119.56%
EUR	.	.	.	.	.
Naive	14.74%	63.83%	0.001	-33.50%	42.07%
Panel C. Subsample 2 (01/1999 - 12/2001)					
AUD	15.83%	58.33%	0.022	-100.42%	98.88%
CAD	10.26%	55.56%	0.091	-100.40%	98.79%
CHF	31.18%	66.67%	0.001	-100.44%	96.99%
DKK	37.42%	69.44%	0.000	-100.41%	98.06%
GBP	-6.18%	47.22%	0.566	-100.43%	99.16%
JPY	32.82%	61.11%	0.004	-100.45%	117.62%
EUR	29.84%	65.71%	0.001	-100.40%	97.79%
Naive	21.48%	63.89%	0.001	-36.79%	54.41%
Panel D: Subsample 3 (01/2002 - 12/2005)					
AUD	26.03%	62.50%	0.002	-100.19%	101.77%
CAD	17.13%	58.33%	0.022	-100.20%	100.94%
CHF	-1.50%	50.00%	0.460	-100.15%	97.15%
DKK	42.22%	70.83%	0.000	-100.15%	100.84%
GBP	17.55%	58.33%	0.022	-100.18%	101.64%
JPY	-0.16%	47.92%	0.669	-100.15%	108.52%
EUR	38.05%	68.75%	0.000	-100.16%	100.87%
Naive	19.90%	72.92%	0.000	-31.54%	39.01%

Notes: In each Panel we report, for the time period indicated in the header, the monthly results of the bilateral options strategies and an equally-weighted (i.e. naive) portfolio of these strategies. $\bar{R}$ denotes the average return, $Hit-Ratio$ the fraction of returns being positive, and $p[Hit-Ratio]$ the probability of achieving the realized or a better hit-ratio just by flipping a coin. Finally, $\bar{R}$ if $R < 0$ states the average of negative returns and $\bar{R}$ if $R > 0$ the average of positive returns.

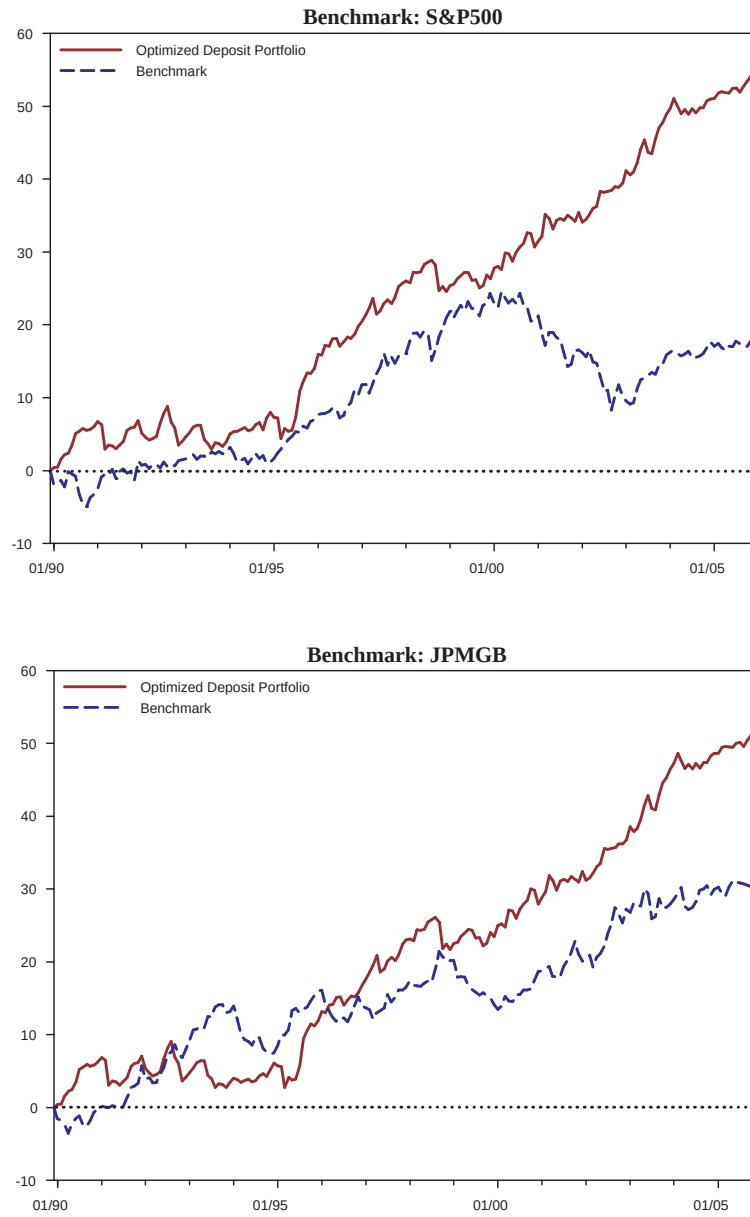
^a Note that for the EUR the evaluation period spans from 01/1999 to 12/2005.

Table 3: Options Strategy Enhanced Portfolios

Benchmark	Strategy	$\bar{R}$	SD	SR	$p[JKM]$	$p[GRS]$	$Jensen's\ \alpha$	SE_{α}
S&P500	Index	10.12%	15.11%	0.40				
	AUD	12.88%	15.58%	0.57	0.0000	0.0002	0.0023	0.0009
	CAD	11.93%	14.85%	0.53	0.0000	0.0016	0.0017	0.0009
	CHF	11.22%	16.16%	0.44	0.0329	0.1382	0.0007	0.0008
	DKK	12.83%	15.65%	0.56	0.0000	0.0003	0.0022	0.0009
	GBP	11.50%	15.13%	0.49	0.0005	0.0126	0.0013	0.0009
	JPY	12.25%	15.09%	0.54	0.0000	0.0008	0.0019	0.0009
	EUR	4.94%	15.25%	0.11	0.0000	.	0.0035	0.0010
	Naive	12.28%	15.09%	0.55	0.0000	0.0007	0.0018	0.0004
	Index	7.66%	14.02%	0.26				
MSCI	AUD	10.45%	14.57%	0.44	0.0000	0.0007	0.0023	0.0009
	CAD	9.50%	13.76%	0.39	0.0000	0.0051	0.0017	0.0009
	CHF	8.79%	14.83%	0.32	0.0050	0.1166	0.0009	0.0009
	DKK	10.40%	14.55%	0.43	0.0000	0.0008	0.0023	0.0009
	GBP	9.07%	14.09%	0.35	0.0002	0.0285	0.0012	0.0009
	JPY	9.82%	13.73%	0.42	0.0000	0.0017	0.0020	0.0009
	EUR	6.19%	14.53%	0.20	0.0000	.	0.0035	0.0010
	Naive	9.85%	13.91%	0.42	0.0000	0.0020	0.0019	0.0004
	Index	7.59%	6.34%	0.56				
	AUD	10.37%	6.94%	0.91	0.0000	0.0000	0.0024	0.0009
JPMGB	CAD	9.43%	7.21%	0.75	0.0000	0.0000	0.0015	0.0009
	CHF	8.72%	6.64%	0.71	0.0000	0.0002	0.0012	0.0009
	DKK	10.33%	7.20%	0.87	0.0000	0.0000	0.0023	0.0009
	GBP	9.00%	7.53%	0.65	0.0000	0.0037	0.0010	0.0009
	JPY	9.75%	6.91%	0.83	0.0000	0.0000	0.0021	0.0010
	EUR	9.89%	7.74%	0.86	0.0000	0.0000	0.0033	0.0010
	Naive	9.77%	6.42%	0.89	0.0000	0.0000	0.0019	0.0004

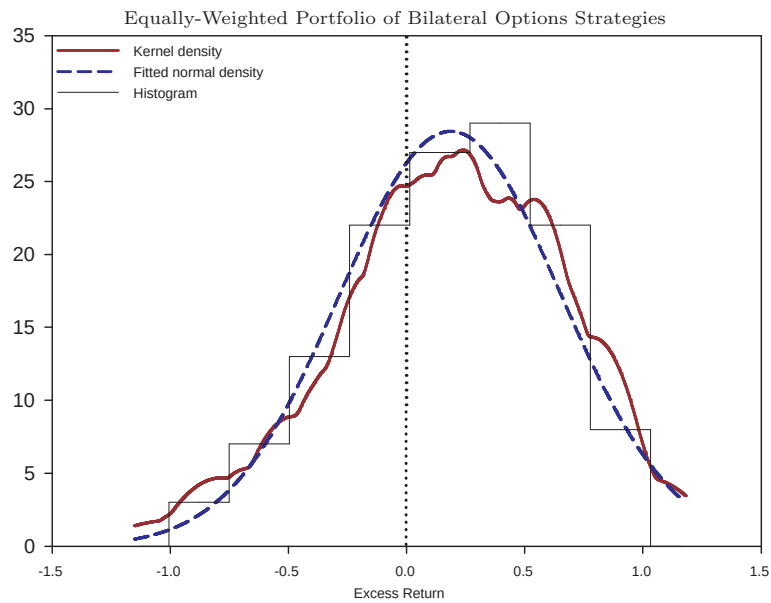
Notes: In this Table we report the annualized results of the options-strategy enhanced portfolio approach, for an evaluation period that spans from 01/1995 to 12/2005. The entries in the first column show the relevant benchmark, i.e. the S&P500, the MSCI, and the JPMGB. Rows labeled *Index* report results of the benchmark and the other rows report results for allocating 1% of funds to the options strategies and the remaining 99% in a benchmark-tracking portfolio. $\bar{R}$ denotes the average return, SD the corresponding standard deviation, and SR the resulting Sharpe ratio. Furthermore, $p[JKM]$ states the significance level at which the strategy's Sharpe ratio is higher than that of the respective benchmark, using Jobson and Korkie (1981) and Memmel (2003). $p[GRS]$ is the significance level at which the strategy has a higher (lower) Sharpe ratio, using Gibbons et al. (1989). Finally, $Jensen's\ \alpha$ denotes the performance contribution measure as in Jensen (1968) (with respect to the respective benchmark) and SE_{α} the corresponding standard error. Note that for the EUR the evaluation period spans from 01/1999 to 12/2005. The benchmark statistics corresponding to the EUR evaluation period are: $\bar{R} = 0.76\%$, $SD = 15.01\%$, $SR = -0.16$ for the S&P500, $\bar{R} = 2.03\%$, $SD = 14.37\%$, $SR = -0.08$ for the MSCI, and $\bar{R} = 5.77\%$, $SD = 6.60\%$, $SR = 0.38$ for the JPMGB.

Figure 1: Cumulative Risk-Adjusted Excess Returns of Optimal Carry-Trade Portfolios



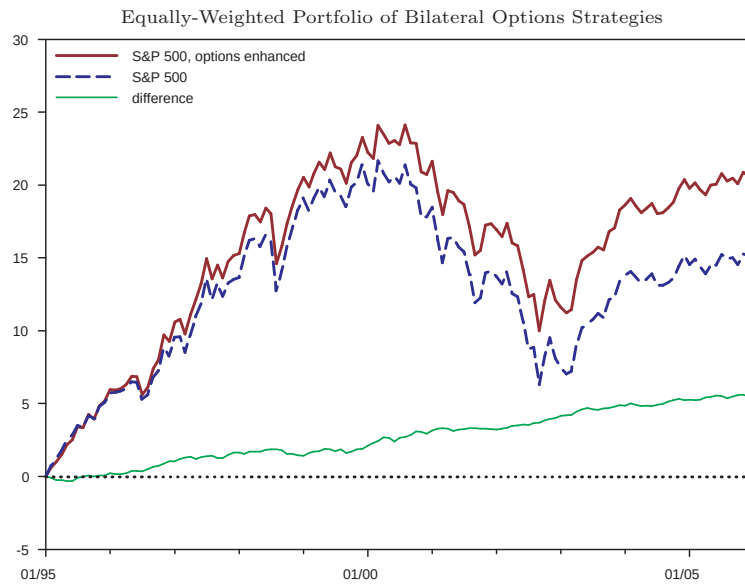
Notes: The Figure shows the cumulative risk-adjusted excess returns (i.e. accumulated monthly excess returns divided by their sample standard deviations) for the optimal carry-trade portfolios using the S&P500 and the JPMGB as benchmarks.

Figure 2: Probability Densities of Options Strategy Excess Returns



Notes: The Figure shows the probability densities of excess returns for an equally-weighted portfolio of bilateral options strategies. The statistics corresponding to the histogram are: mean = 0.1848, standard deviation = 0.6718, skewness = -0.3495, kurtosis = 2.7878, and Jarque-Bera test statistic = 2.91.

Figure 3: Cumulative Risk-Adjusted Excess Returns of Option-Enhanced S&P500 Investment



Notes: The Figure shows the cumulative risk-adjusted excess returns (i.e. accumulated monthly excess returns divided by their sample standard deviations) for allocating 1% of funds to an equally-weighted portfolio of bilateral options strategies and the remaining 99% in a portfolio tracking the S&P500.